

EECS 570

Lecture 3

Shared-Memory and Synchronization

Fall 2025

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<http://www.eecs.umich.edu/courses/eecs570/>

Slides developed in part by Profs. Adve, Falsafi, Martin, Roth, Nowatzky, and Wenisch of EPFL, CMU, UPenn, U-M, UIUC.



Announcements

Discussion this Friday:

Follow-up on Programming Assignment 1

Readings

Using Message Passing to Transfer Data Between Threads -
The Rust Programming Language

Michael Scott, Shared-Memory Synchronization Synthesis
Lectures on Computer Architecture (Ch. 1, 4.0-4.3.3,
5.0-5.2.5)

Agenda

Shared-Memory programming model

Brief intro to architecture support

Synchronization operations

- Locks
- Barriers

Shared Memory Programming Model

Shared-Memory Model

Execution Contexts: Share a single address space

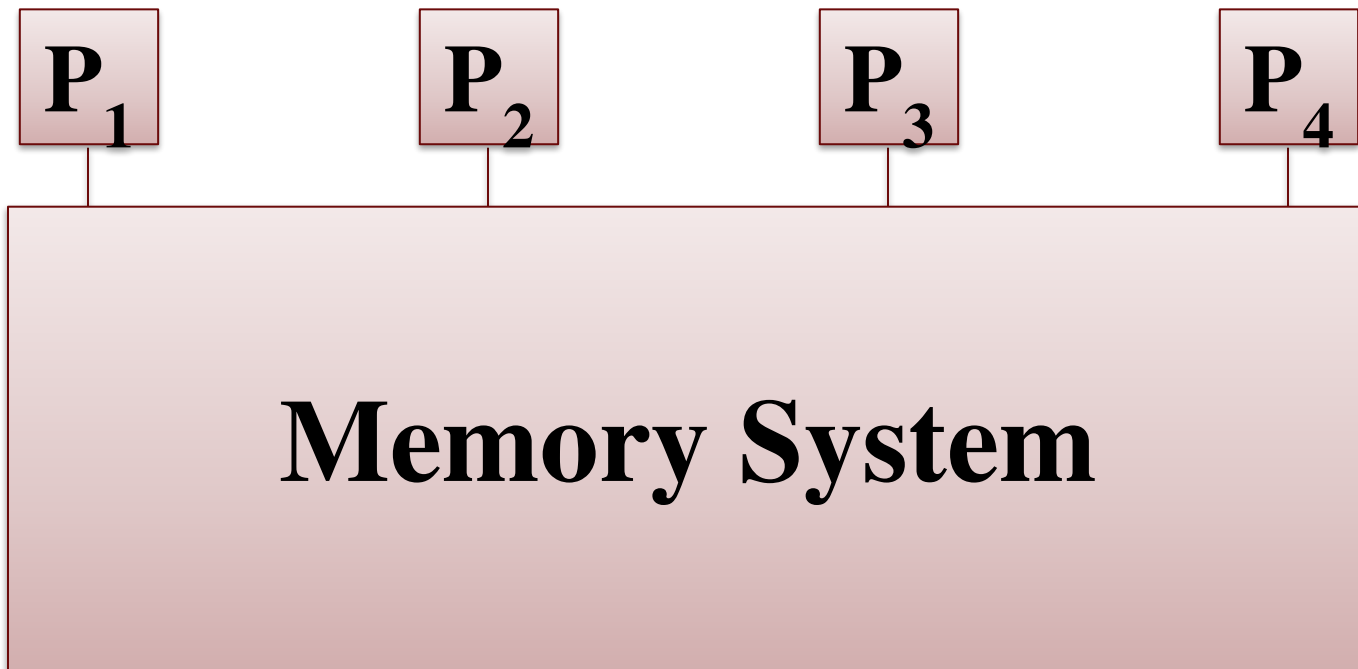
Models:

MIMD: Multiple programs

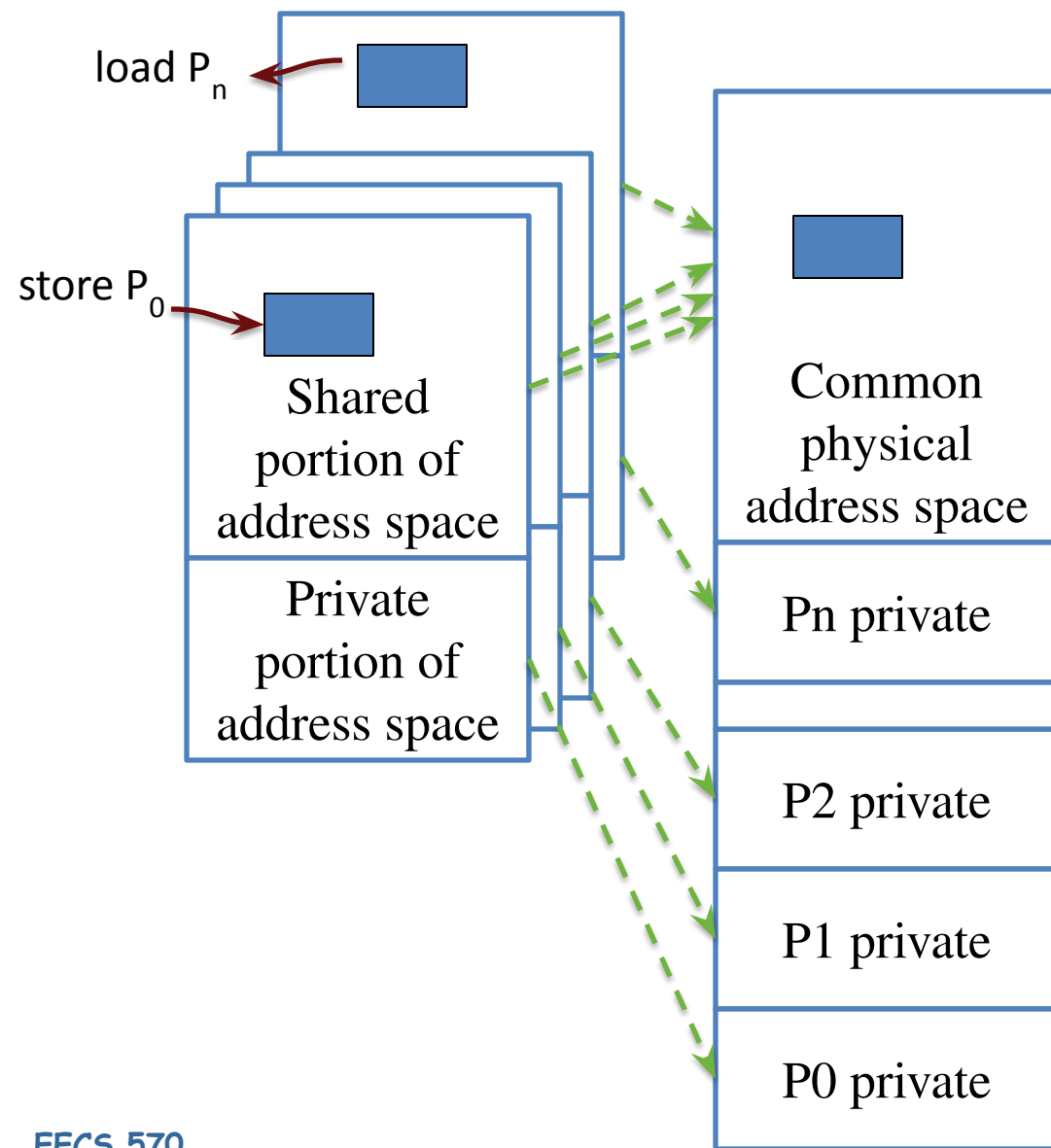
SPMD: Multiple copies of one program

Communication: Implicit via loads/stores

Theory: Based on PRAM model



Global Shared Physical Address Space



- Communication, sharing, synchronization via loads/stores to shared variables
- Facilities for address translation between local/global address spaces
- Requires OS support to maintain this mapping

Why Shared Memory?

Pluses:

- Intuitive for programmers – no need for explicit comm.
- OS needs minimal evolutionary extensions
- Simplifies communication without OS

Minuses:

- Easier to make mistakes with synchronization
- Implicit communication makes optimization harder
- Needs complex hardware support for comm. (e.g., coherence)

• Result:

Shared-memory multi-core and GPUs are common today

Thread-Level Parallelism

```
struct acct_t { int bal; };  
shared struct acct_t  accts[MAX_ACCT];  
int id,amt;  
if (accts[id].bal >= amt)  
{  
    accts[id].bal -= amt;  
    spew_cash();  
}
```

0: addi r1,accts,r3
1: ld 0(r3),r4
2: blt r4,r2,6
3: sub r4,r2,r4
4: st r4,0(r3)
5: call spew_cash

- **Thread-level parallelism (TLP)**

- r Collection of asynchronous tasks: not started and stopped together
- r Data shared loosely, dynamically

- Example: database/web server (each query is a thread)

- r **accts** is **shared**, can't register allocate even if it were scalar
- r **id** and **amt** are private variables, register allocated to **r1**, **r2**

Synchronization

- Mutual exclusion : locks, ...
- Order : barriers, signal-wait, ...
- Implemented using read/write/RMW to shared location
 - r Language-level:
 - m libraries (e.g., locks in pthread)
 - m Programmers can write custom synchronizations
 - r Hardware ISA
 - m E.g., test-and-set
- OS provides support for managing threads
 - r scheduling, fork, join, futex signal/wait

Cache Coherence

Processor 0

0: `addi r1,accts,r3`

1: `ld 0(r3),r4`

2: `blt r4,r2,6`

3: `sub r4,r2,r4`

4: `st r4,0(r3)`

5: `call spew_cash`

Processor 1

0: `addi r1,accts,r3`

1: `ld 0(r3),r4`

2: `blt r4,r2,6`

3: `sub r4,r2,r4`

4: `st r4,0(r3)`

5: `call spew_cash`



- Two \$100 withdrawals from account #241 at two ATMs
 - r Each transaction maps to thread on different processor
 - r Track `accts[241].bal` (address is in `r3`)

No-Cache, No-Problem

Processor 0

0: `addi r1,accts,r3`

1: `ld 0(r3),r4`

2: `blt r4,r2,6`

3: `sub r4,r2,r4`

4: `st r4,0(r3)`

5: `call spew_cash`

Processor 1

0: `addi r1,accts,r3`

1: `ld 0(r3),r4`

2: `blt r4,r2,6`

3: `sub r4,r2,r4`

4: `st r4,0(r3)`

5: `call spew_cash`

		500
		500

		400
--	--	-----

		400
--	--	-----

		300
--	--	-----

- Scenario I: processors have no caches
 - r No problem

Cache Incoherence

Processor 0

0: addi r1,accts,r3

1: ld 0(r3),r4

2: blt r4,r2,6

3: sub r4,r2,r4

4: st r4,0(r3)

5: call spew_cash

Processor 1

0: addi r1,accts,r3

1: ld 0(r3),r4

2: blt r4,r2,6

3: sub r4,r2,r4

4: st r4,0(r3)

5: call spew_cash

		500
V:500		500

D:400		500
-------	--	-----

D:400	V:500	500
-------	-------	-----

D:400	D:400	500
-------	-------	-----

- Scenario II: processors have write-back caches
 - r Potentially 3 copies of **accts[241].bal**: memory, p0\$, p1\$
 - r Can get incoherent (out of sync)

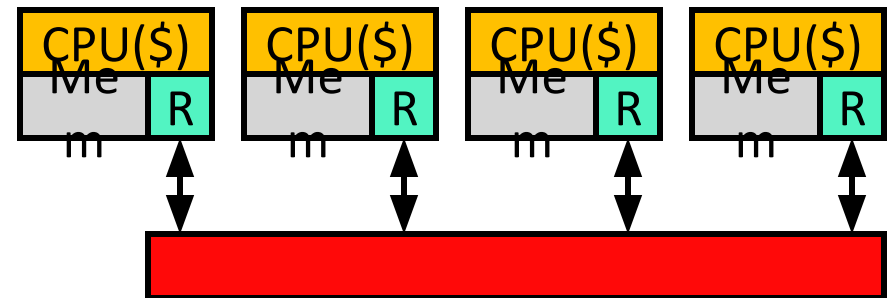
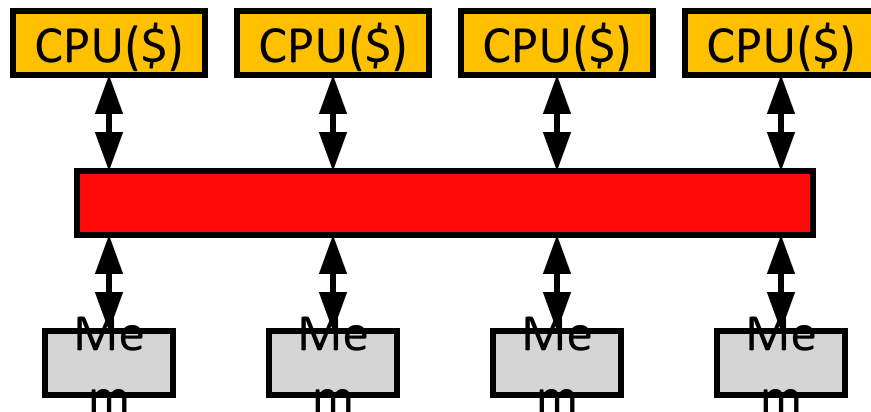
Paired vs. Separate Processor/Memory?

- **Separate processor/memory**

- **Uniform memory access (UMA):** equal latency to all memory
 - + Simple software, doesn't matter where you put data
 - Lower peak performance
- Bus-based UMAs common: **symmetric multi-processors (SMP)**

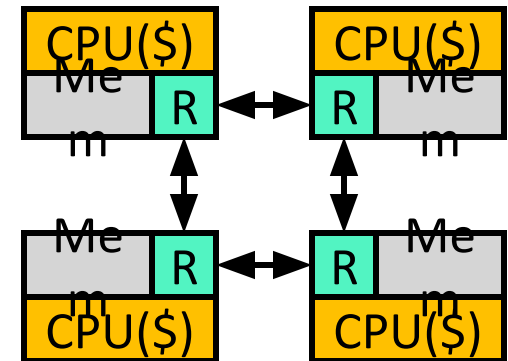
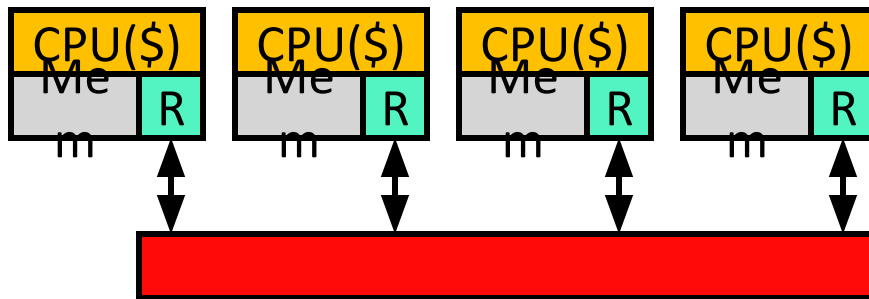
- **Paired processor/memory**

- **Non-uniform memory access (NUMA):** faster to local memory
 - More complex software: where you put data matters
 - + Higher peak performance: assuming proper data placement

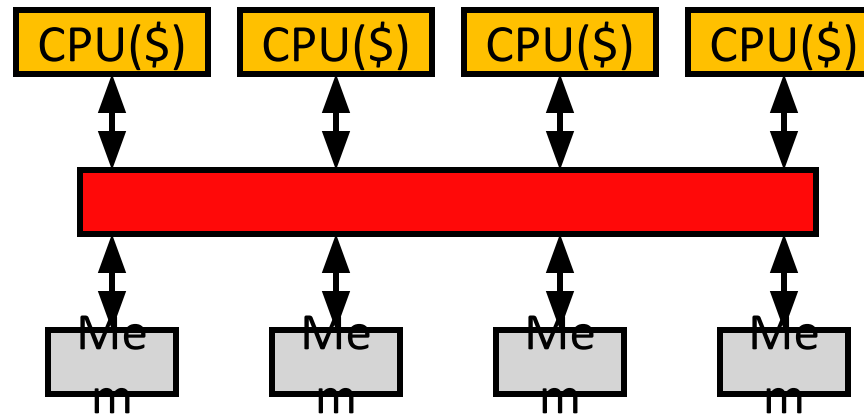


Shared vs. Point-to-Point Networks

- **Shared network:** e.g., bus (left)
 - + Low latency
 - Low bandwidth: doesn't scale beyond ~16 processors
 - + Shared property simplifies cache coherence protocols (later)
- **Point-to-point network:** e.g., mesh or ring (right)
 - Longer latency: may need multiple “hops” to communicate
 - + Higher bandwidth: scales to 1000s of processors
 - Cache coherence protocols are complex



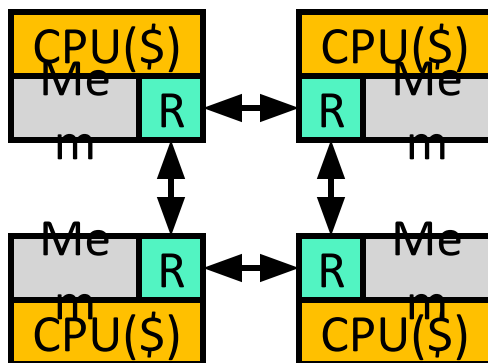
Implementation #1: Snooping Bus MP



Bus-based systems

- r Typically small: 2–8 (maybe 16) processors
- r Typically, processors split from memories (UMA)
 - m **Multiple processors (cores) on single chip (multi-core)**

Implementation #2: Scalable MP



- General point-to-point network-based systems
 - r Typically, processor/memory/router blocks (NUMA)
 - m **Glueless MP**: no need for additional “glue” chips
 - r Can be arbitrarily large: 1000’s of processors
 - m **Massively parallel processors (MPPs)**
 - r AMD Infinity Fabric, Intel UPI
 - r nVidia’s NVLink (scales to 10s of GPUs)

Snooping Cache-Coherence Protocols

Bus provides serialization point

Each cache controller “snoops” all bus transactions

- r take action to ensure coherence
 - m invalidate
 - m update
 - m supply value
- r depends on state of the block and the protocol

Scalable Cache Coherence

- **Scalable cache coherence**: two part solution
- Part I: **bus bandwidth**
 - r Replace non-scalable bandwidth substrate (bus)...
 - r ...with scalable bandwidth one (point-to-point network, e.g., mesh)
- Part II: **processor snooping bandwidth**
 - r Interesting: most snoops result in no action
 - r Replace non-scalable broadcast protocol (spam everyone)...
 - r ...with scalable **directory protocol** (only spam processors that care)
- We will cover this in Unit 2

Shared Memory Summary

- Shared-memory multiprocessors
 - + “Simple” software: easy data sharing, handles both DLP & TLP
 - ...but hard to get fully correct!
 - Complex hardware: must provide illusion of global address space
- Two basic implementations
 - r **Symmetric (UMA) multi-processors (SMPs)**
 - m Underlying communication network: bus (ordered)
 - + Low-latency, simple protocols
 - Low-bandwidth, poor scalability
 - r **Scalable (NUMA) multi-processors (MPPs)**
 - m Underlying communication network: point-to-point (often unordered)
 - + Scalable bandwidth

Synchronization

Synchronization objectives

- Low overhead
 - r Synchronization can limit scalability (E.g., single-lock OS kernels)
- Correctness (and ease of programmability)
 - r Synchronization failures are extremely difficult to debug
- Coordination of HW and SW
 - r SW semantics must be tightly specified to prove correctness
 - r HW can often improve efficiency

Synchronization Forms

- Mutual exclusion (critical sections)
 - r Lock & Unlock
- Event Notification
 - r Point-to-point (producer-consumer, flags)
 - r I/O, interrupts, exceptions
- Barrier Synchronization
- Higher-level constructs
 - r Queues, software pipelines, (virtual) time, counters
- Novel research solution: optimistic concurrency control

Anatomy of a Synchronization Op

- Acquire Method

- r Way to obtain the lock or proceed past the barrier

- Waiting Algorithm

- r Spin (aka busy wait)
 - m Waiting process repeatedly tests a location until it changes
 - m Releasing process sets the location
 - m Lower overhead, but wastes CPU resources
 - m Can cause interconnect traffic
 - r Block (aka suspend)
 - m Waiting process is descheduled
 - m High overhead, but frees CPU to do other things
 - r Hybrids (e.g., spin, then block)

- Release Method

Way to allow other processes to proceed

HW/SW Implementation Trade-offs

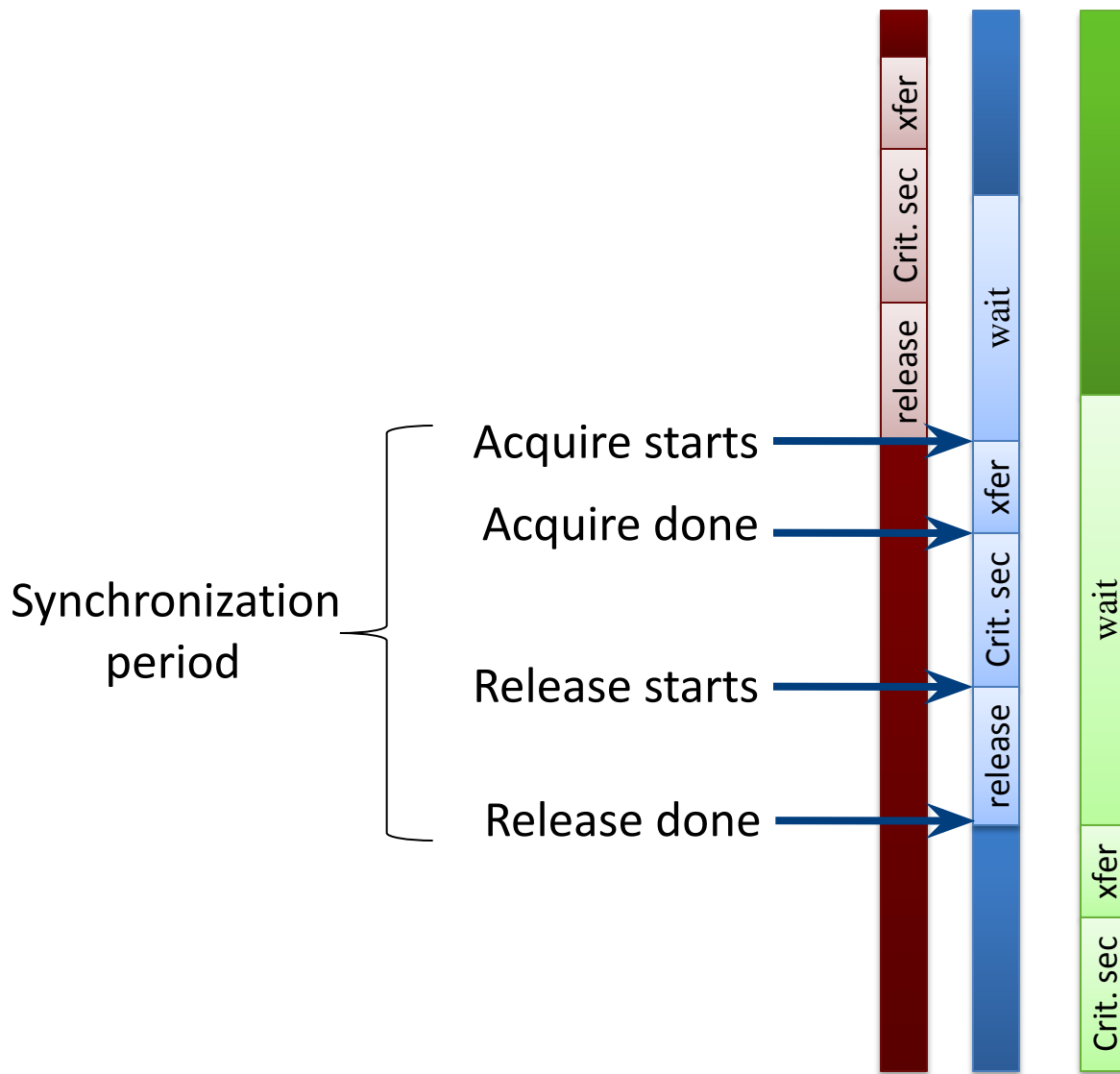
- User wants high-level (ease of programming)
 - r `LOCK(lock_variable); UNLOCK(lock_variable)`
 - r `BARRIER(barrier_variable, numprocs)`
- SW advantages: flexibility, portability
- HW advantages: speed
- Design objectives:
 - r Low latency
 - r Low traffic
 - r Low storage
 - r Scalability (“wait-free”-ness)
 - r Fairness

Challenges

- Same sync may have different behavior at different times
 - r Lock accessed with low or high contention
 - r Different performance needs: low latency vs. high throughput
 - r Different algorithms best for each, need different primitives
- Multiprogramming can change sync behavior
 - r Process scheduling or other resource interactions
 - r May need algorithms that are worse in dedicated case
- Rich area of SW/HW interactions
 - r Which primitives are available?
 - r What communication patterns cost more/less?

Locks

Lock-based Mutual Exclusion



No contention:

- Want low latency

Contention:

- Want low period
- Low traffic
- Fairness

How Not to Implement Locks

LOCK

```
while (lock_variable == 1);
```

```
lock_variable = 1;
```



Context switch!

UNLOCK

```
lock_variable = 0;
```

Solution: Atomic Read-Modify-Write

- Test&Set(r,x)

`{ r=m[x] ; m[x]=1 ; }`

- r is register

- m[x] is memory location x

- Fetch&Op(r1,r2,x,op)

`{ r1=m[x] ; m[x]=op(r1,r2) ; }`

- Swap(r,x)

`{ temp=m[x] ; m[x]=r ; r=temp ; }`

- Compare&Swap(r1,r2,x)

`{ temp=r2 ; r2=m[x] ; if r1==r2 then m[x]=temp ; }`

Implementing RMWs

- Bus-based systems:
 - r Hold bus and issue load/store operations without any intervening accesses by other processors
- Perform operation at shared point in the memory hierarchy
 - r E.g., if L1s are private and L2 is shared, perform sync ops at L2
 - m Need to invalidate lines for the address in the private L1s!
- Scalable systems
 - r Acquire exclusive ownership via cache coherence
 - r Perform load/store operations without allowing external coherence requests

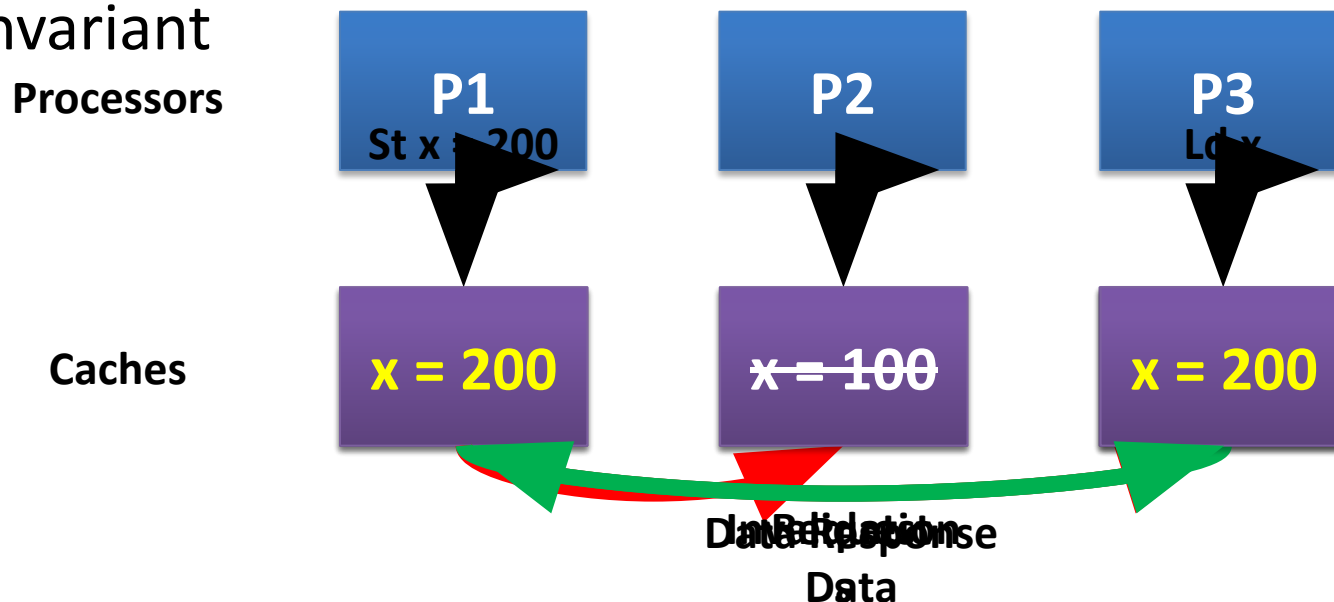
Load-Locked Store-Conditional

- Load-locked
 - r Issues a normal load...
 - r ...and sets a **flag** and **address** field
- Store-conditional
 - r Checks that **flag** is set and **address** matches...
 - r ...only then performs store
- **Flag** is cleared by
 - r Invalidation
 - r Cache eviction
 - r Context switch

```
lock: while (1) {  
    load-locked r1, lock_variable  
    if (r1 == 0) {  
        mov r2 = 1  
        if (SC r2, lock_variable) break;  
    }  
}  
  
unlock: st lock_variable, #0
```

Coherence Protocol Example

- If P1 updates the value of x to 200, the stale value of x in other processors must be **invalidated**
- If P3 wants to subsequently read/write x, it must request the new value
- **SWMR** = Single-Writer Multiple Readers, **DVI** = Data Value Invariant



Test-and-Set Spin Lock (T&S)

- Lock is “acquire”, Unlock is “release”

- `acquire(lock_ptr) :`

```
    while (true) :
```

```
        // Perform "test-and-set"
```

```
        // UNLOCKED = 0, LOCKED = 1
```

```
        test_and_set(old, lock_ptr)
```

```
        if (old == UNLOCKED) :
```

```
            break    // lock acquired!
```

```
        // keep spinning, back to top of while loop
```

- `release(lock_ptr) :`

```
    store[lock_ptr] <- UNLOCKED
```

- Performance problem

Test-and-Test-and-Set Spin Lock (TTS)

- `acquire(lock_ptr) :`

- `while (true) :`

- // Perform "test"*

- `load [lock_ptr] -> original_value`

- `if (original_value == UNLOCKED) :`

- // Perform "test-and-set"*

- `test_and_set(old, lock_ptr)`

- `if (old == UNLOCKED) :`

- `break     // lock acquired!`

- // keep spinning, back to top of while loop*

- `release(lock_ptr) :`

- `store[lock_ptr] <- UNLOCKED`

- **Now “spinning” is read-only, on local cached copy**

TTS Lock Performance Issues

- **Performance issues remain**

- r Every time the lock is released...
 - m All spinning cores get invalidated -> lots of coherence traffic
 - m All spinning cores would then load the lock addr to keep spinning, and likely try to T&S the block
 - q More coherence traffic!
- r Causes a storm of coherence traffic, clogs things up badly

- **One solution: backoff**

- r Instead of spinning constantly, check less frequently
- r Exponential backoff works well in practice

- **Another problem with spinning**

- r Processors can spin really fast, starve threads on the same core!
- r Solution: x86 adds a “PAUSE” instruction
 - m Tells processor to suspend the thread for a short time

Ticket Locks

- **To ensure fairness and reduce coherence storms**

- Locks have two counters: `next_ticket`, `now_serving`

- `r` Deli counter

- `acquire(lock_ptr) :`

- `r my_ticket = fetch_and_increment(lock_ptr->next_ticket)`

- `r while(lock_ptr->now_serving != my_ticket); // spin`

- `release(lock_ptr) :`

- `r lock_ptr->now_serving = lock_ptr->now_serving + 1`

- `m` (Just a normal store, not an atomic operation, why?)

- Summary of operation

- `r` To “get in line” to acquire the lock, CAS on `next_ticket`

- `r` Spin on `now_serving`

Ticket Locks

- **Properties**

- r Less of a “thundering herd” coherence storm problem
 - m To acquire, only need to read new value of `now_serving`
- r No CAS on critical path of lock handoff
 - m Just a non-atomic store
- r FIFO order (fair)
 - m Good, but only if the O.S. hasn’t swapped out any threads!

- **Padding**

- r Allocate `now_serving` and `next_ticket` on different cache blocks
 - m `struct { int now_serving; char pad[60]; int next_ticket; } ...`
- r Two locations reduces interference

- **Proportional backoff**

- r Estimate of wait time: $(\text{my_ticket} - \text{now_serving}) * \text{average hold time}$

Array-Based Queue Locks

- **Why not give each waiter its own location to spin on?**
 - r Avoid coherence storms altogether!
- **Idea: “slot” array of size N: “go ahead” or “must wait”**
 - m Initialize first slot to “go ahead”, all others to “must wait”
 - m Padded one slot per cache block,
 - r Keep a “next slot” counter (similar to “next_ticket” counter)
- Acquire: “get in line”
 - r $\text{my_slot} = (\text{atomic increment of “next slot” counter}) \bmod N$
 - r Spin while `slots[my_slot]` contains “must_wait”
 - r Reset `slots[my_slot]` to “must wait”
- Release: “unblock next in line”
 - r Set `slots[my_slot+1 mod N]` to “go ahead”

Array-Based Queue Locks

- Variants: Anderson 1990, Graunke and Thakkar 1990
 - Desirable properties
 - r Threads spin on dedicated location
 - m Just two coherence misses per handoff
 - m Traffic independent of number of waiters
 - r FIFO & fair (same as ticket lock)
 - Undesirable properties
 - r Higher uncontended overhead than a TTS lock
 - r Storage $O(N)$ for each lock
 - m 128 threads at 64B padding: 8KBs per lock!
 - m What if N isn't known at start?
 - List-based locks address the $O(N)$ storage problem
- EECS 570r Several variants of list-based locks: MCS 1991, CLH 1993/1994

List-Based Queue Lock (MCS)

- A “lock” is a pointer to a linked list node

- r next node pointer

- r boolean must_wait

- r Each thread has its own local pointer to a node “I”

- acquire(lock) :

```
I->next = null;
```

```
predecessor = fetch_and_store(lock, I)
```

```
if predecessor != nil           //some node holds lock
```

```
    I->must_wait = true
```

```
    predecessor->next = I        //predecessor must wake us
```

```
    repeat while I->must_wait    //spin till lock is free
```

- release(lock) :

```
if (I->next == null)           //no known successor
```

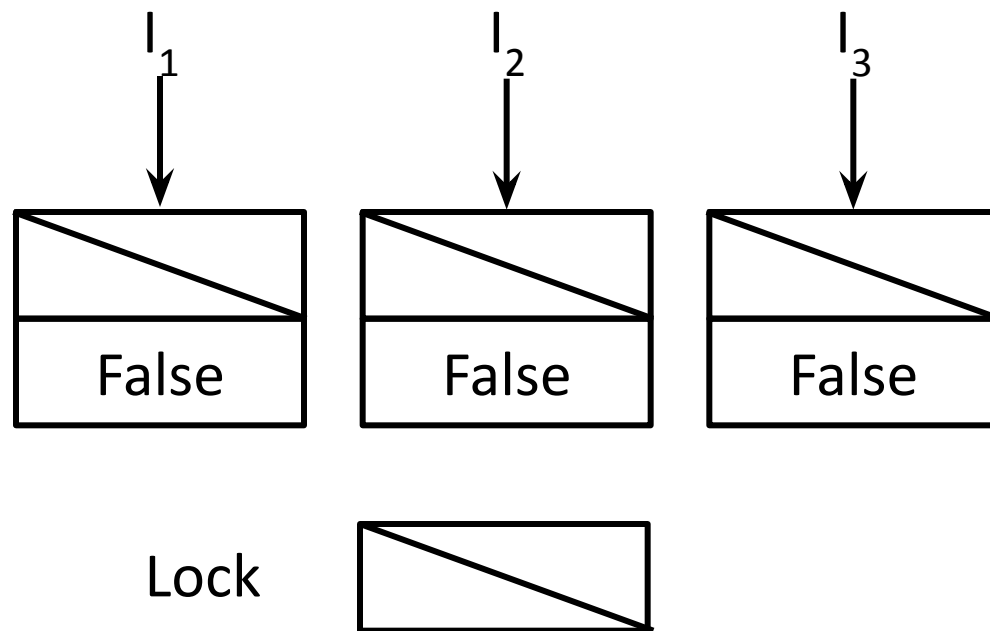
```
    if compare_and_swap(lock, I, nil) //make sure...
```

```
        return                //CAS succeeded; lock freed
```

```
    repeat while I->next = nil    //spin to learn successor
```

```
    I->next->must_wait = false    //wake successor
```

MCS Lock Example: Time 0



- `acquire(lock) :`

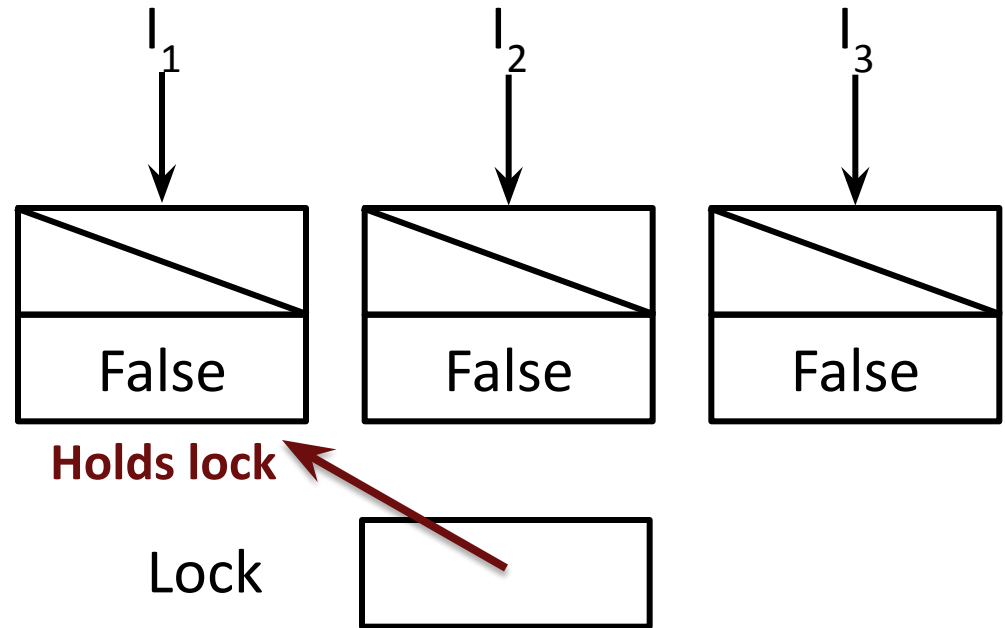
```
I->next = null;  
pred = FAS(lock, I)  
if pred != nil  
    I->must_wait = true  
    pred->next = I  
    repeat while I->must_wait
```

- `release(lock) :`

```
if (I->next == null)  
    if CAS(lock, I, nil)  
        return  
    repeat while I->next == nil  
    I->next->must_wait = false
```

MCS Lock Example: Time 1

- t_1 : Acquire(L)



- `acquire(lock) :`

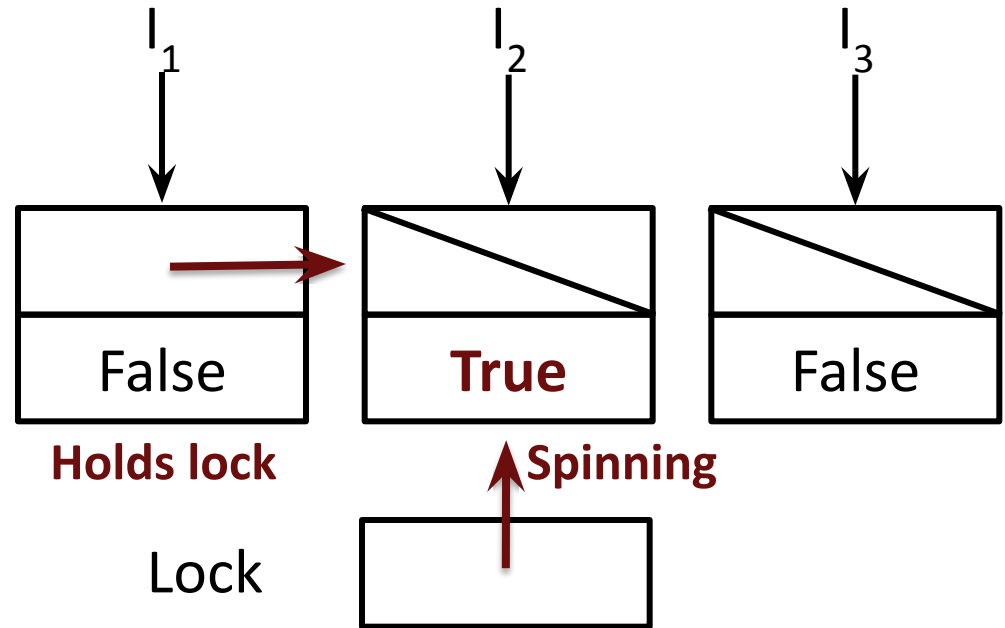
```
I->next = null;  
pred = FAS(lock, I)  
if pred != nil  
    I->must_wait = true  
    pred->next = I  
    repeat while I->must_wait
```

- `release(lock) :`

```
if (I->next == null)  
    if CAS(lock, I, nil)  
        return  
    repeat while I->next == nil  
    I->next->must_wait = false
```

MCS Lock Example: Time 2

- t_1 : Acquire(L)
- t_2 : Acquire(L)



- `acquire(lock) :`

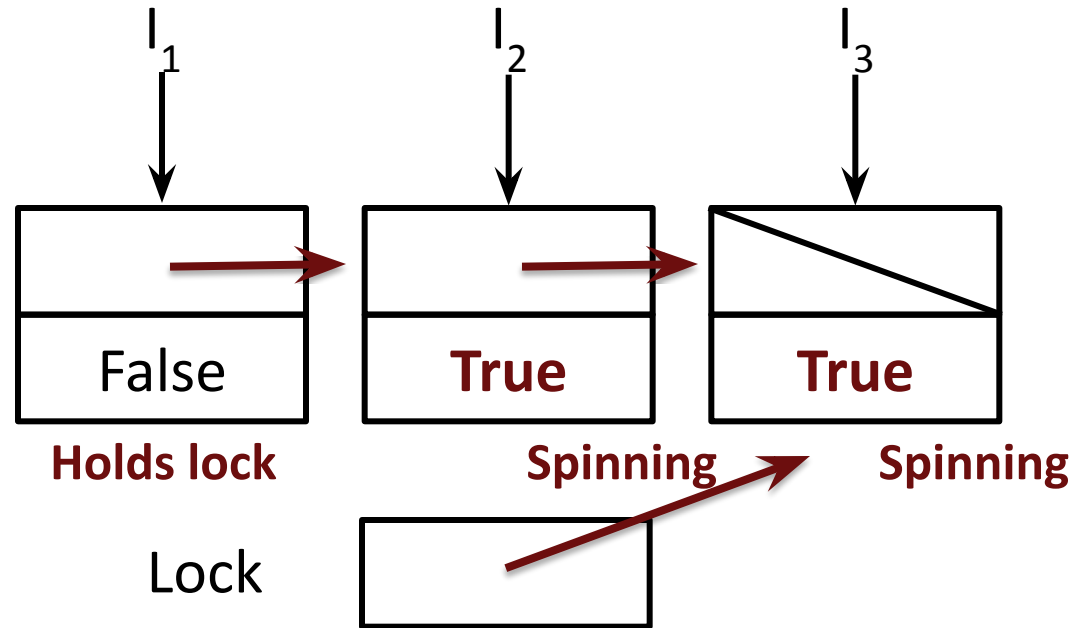
```
I->next = null;  
pred = FAS(lock, I)  
if pred != nil  
    I->must_wait = true  
    pred->next = I  
    repeat while I->must_wait
```

- `release(lock) :`

```
if (I->next == null)  
    if CAS(lock, I, nil)  
        return  
    repeat while I->next == nil  
I->next->must_wait = false
```

MCS Lock Example: Time 3

- t_1 : Acquire(L)
- t_2 : Acquire(L)
- t_3 : Acquire(L)



- `acquire(lock) :`

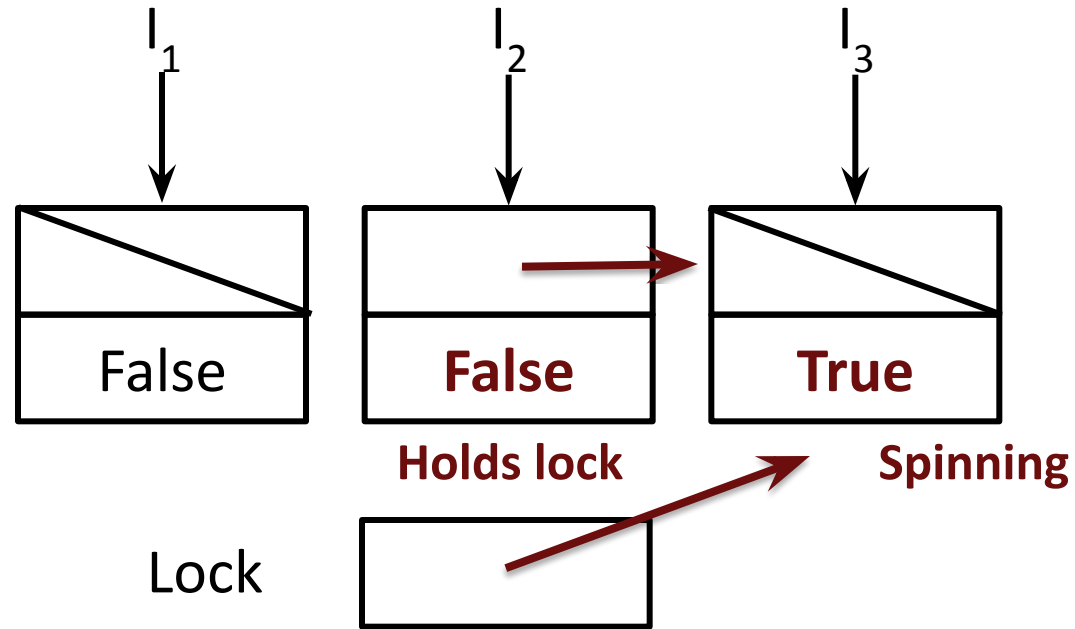
```
I->next = null;  
pred = FAS(lock, I)  
if pred != nil  
    I->must_wait = true  
    pred->next = I  
    repeat while I->must_wait
```

- `release(lock) :`

```
if (I->next == null)  
    if CAS(lock, I, nil)  
        return  
    repeat while I->next == nil  
    I->next->must_wait = false
```

MCS Lock Example: Time 4

- t_1 : Acquire(L)
- t_2 : Acquire(L)
- t_3 : Acquire(L)
- t_1 : Release(L)



- `acquire(lock) :`

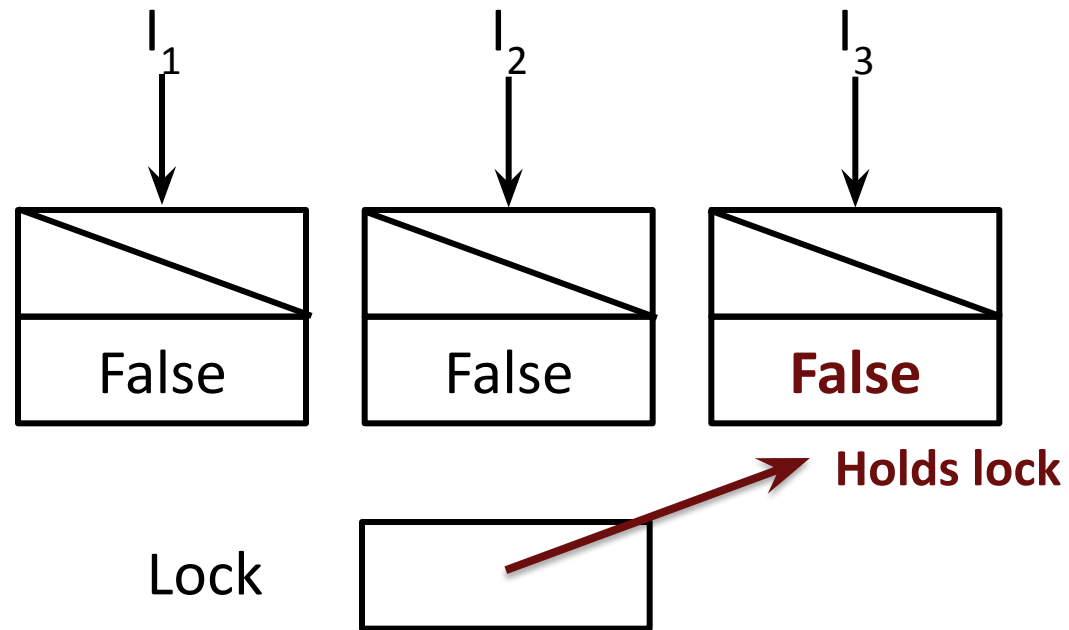
```
I->next = null;  
pred = FAS(lock, I)  
if pred != nil  
    I->must_wait = true  
    pred->next = I  
    repeat while I->must_wait
```

- `release(lock) :`

```
if (I->next == null)  
    if CAS(lock, I, nil)  
        return  
    repeat while I->next == nil  
I->next->must_wait = false
```

MCS Lock Example: Time 5

- t_1 : Acquire(L)
- t_2 : Acquire(L)
- t_3 : Acquire(L)
- t_1 : Release(L)
- t_2 : Release(L)



- `acquire(lock) :`

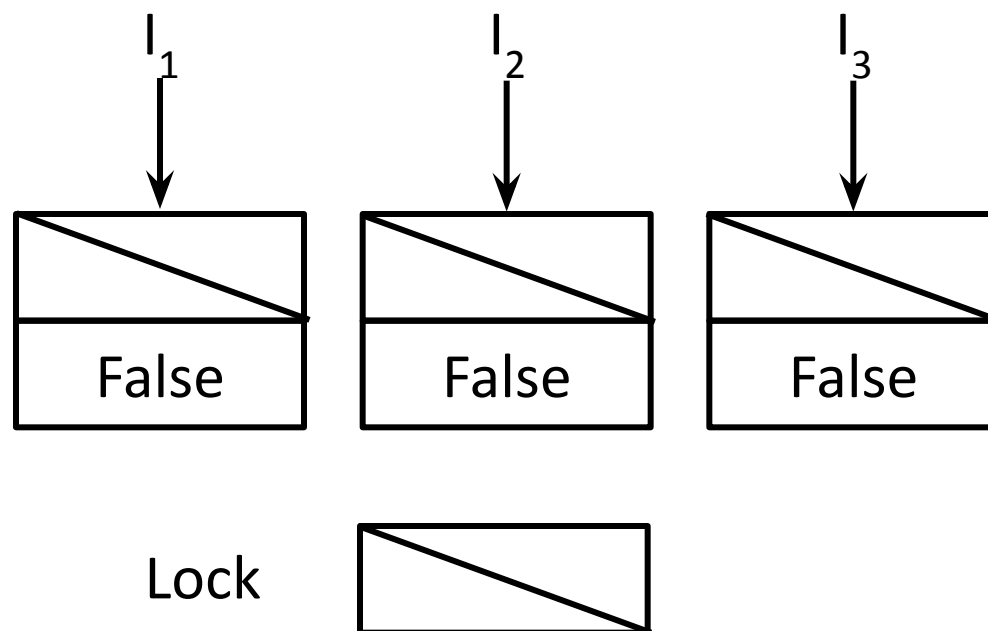
```
I->next = null;
pred = FAS(lock, I)
if pred != nil
    I->must_wait = true
    pred->next = I
    repeat while I->must_wait
```

- `release(lock) :`

```
if (I->next == null)
    if CAS(lock, I, nil)
        return
    repeat while I->next == nil
    I->next->must_wait = false
```

MCS Lock Example: Time 6

- t_1 : Acquire(L)
- t_2 : Acquire(L)
- t_3 : Acquire(L)
- t_1 : Release(L)
- t_2 : Release(L)
- t_3 : Release(L)



• acquire(lock) :

```

I->next = null;
pred = FAS(lock, I)
if pred != nil
    I->must_wait = true
    pred->next = I
    repeat while I->must_wait
    
```

• release(lock) :

```

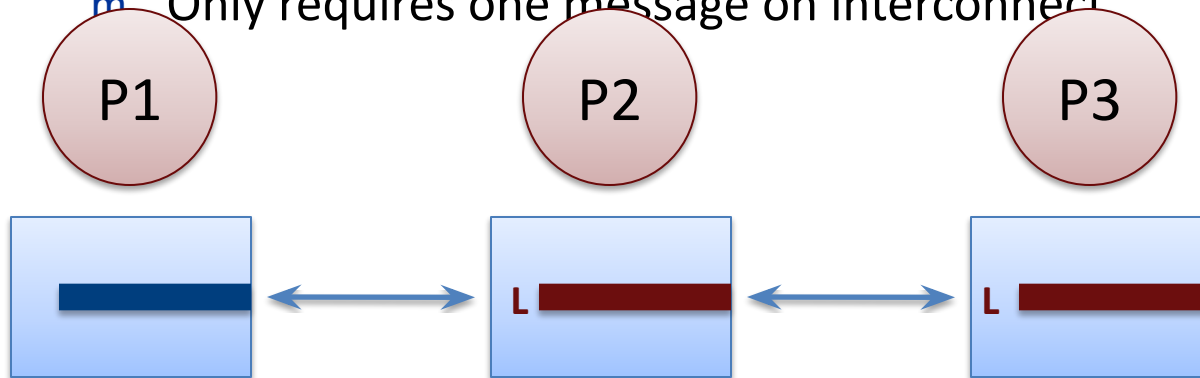
if (I->next == null)
    if CAS(lock, I, nil)
        return
    repeat while I->next == nil
    I->next->must_wait = false
    
```

release() w/o CAS is more complex; see paper

Queue-based locks in HW: QOLB

- **Queue On Lock Bit**

- r HW maintains doubly-linked list between requesters
 - m This is a key idea of “Scalable Coherence Interface”, see Unit 3
- r Augment cache with “locked” bit
 - m Waiting caches spin on local “locked” cache line
- r Upon release, lock holder sends line to 1st requester
 - m Only requires one message on interconnect



Fundamental Mechanisms to Reduce Overheads

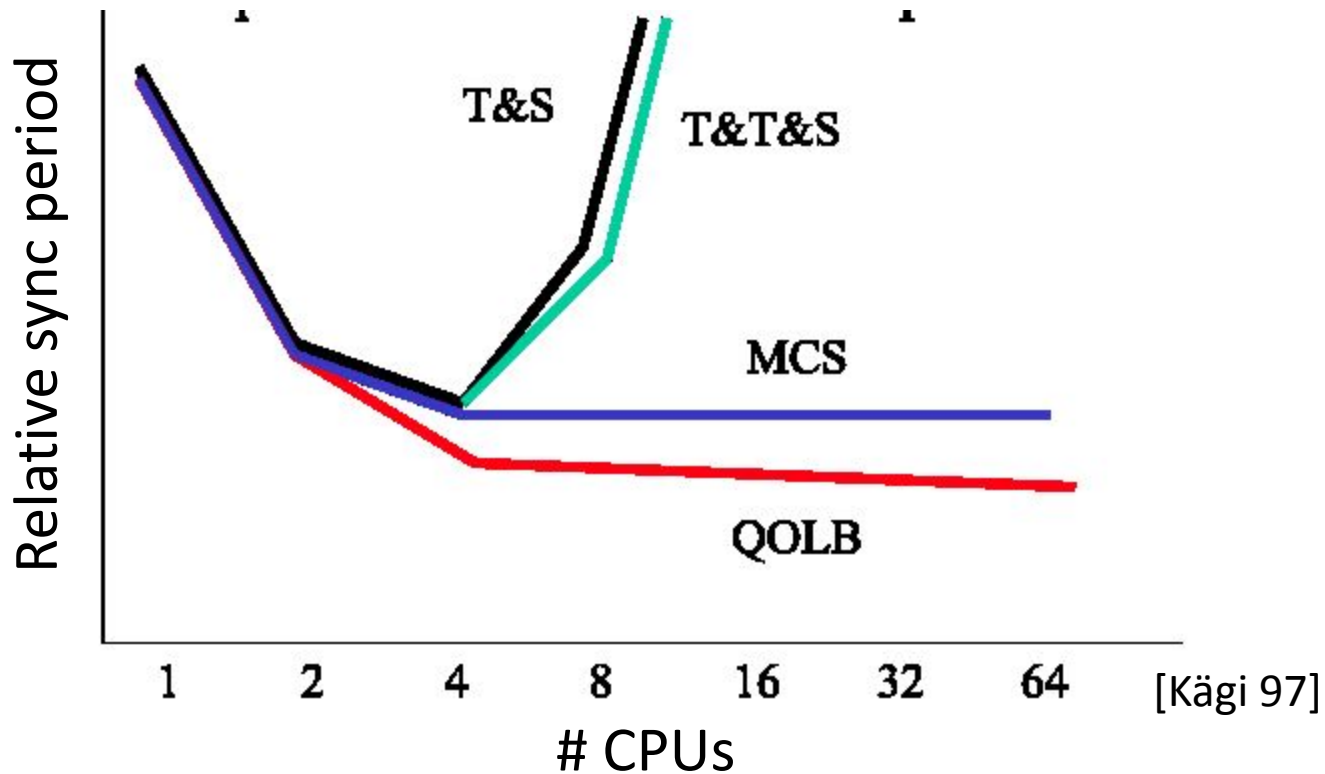
[Kägi, Burger, Goodman ASPLOS 97]

- **Basic mechanisms**

- r Local Spinning
- r Queue-based locking
- r Collocation
- r Synchronous Prefetch

	Local Spin	Queue	Collocation	Prefetch
T&S	No	No	Optional	No
T&T&S	Yes	No	Optional	No
MCS	Yes	Yes	Partial	No
QOLB	yes	Yes	Optional	Yes

Microbenchmark Analysis



Performance of Locks

- **Contention vs. No Contention**

- r Test-and-Set best when no contention
- r Queue-based is best with medium contention
- r Idea: switch implementation based on lock behavior
 - m Reactive Synchronization – Lim & Agarwal 1994
 - m SmartLocks – Eastep et al 2009

- **High-contention indicates poorly written program**

- r Need better algorithm or data structures

Point-to-Point Event Synchronization

- Can use normal variables as flags

```
a = f(x);           while (flag == 0);  
flag = 1;           b = g(a);
```

- If we know initial conditions

```
a = f(x);           while (a == 0);  
                      b = g(a);
```

- **Assumes Sequential Consistency!**

- Full/Empty Bits

- r Set on write
 - r Cleared on read
 - r Can't write if set, can't read if clear

Barriers

Barriers

- Physics simulation computation

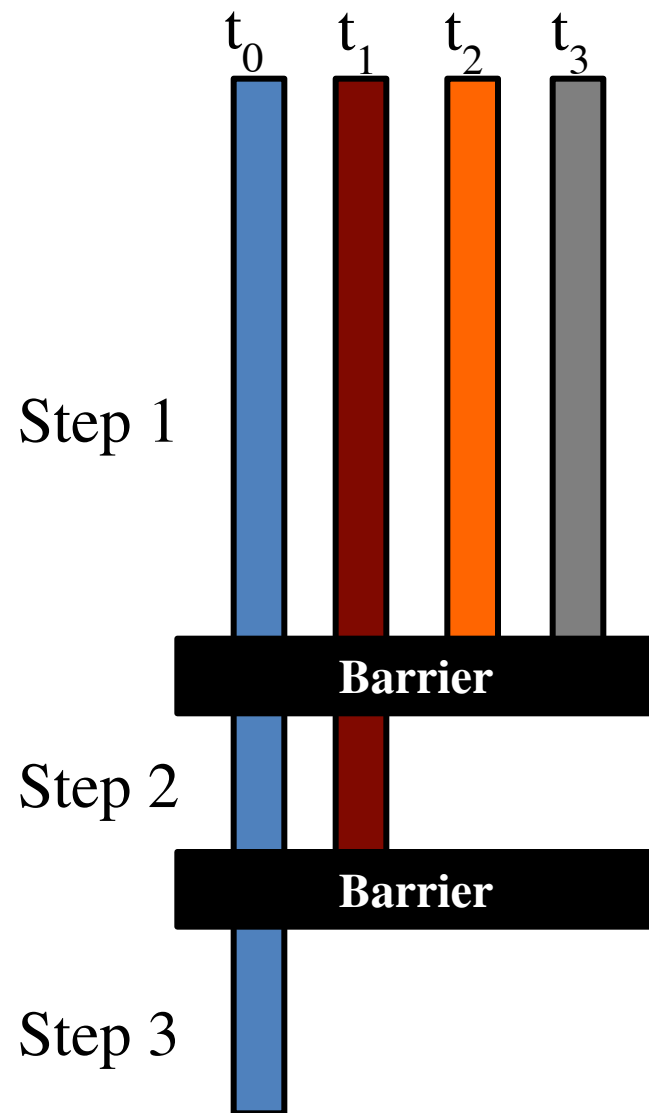
- r Divide up each timestep computation into N independent pieces
 - r Each timestep: compute independently, synchronize

- Example: each thread executes:

```
segment_size = total_particles / number_of_threads
my_start_particle = thread_id * segment_size
my_end_particle = my_start_particle + segment_size - 1
for (timestep = 0; timestep += delta; timestep < stop_time):
    calculate_forces(t, my_start_particle, my_end_particle)
    barrier()
    update_locations(t, my_start_particle, my_end_particle)
    barrier()
```

- Barrier? All threads wait until all threads have reached it

Example: Barrier-Based Merge Sort



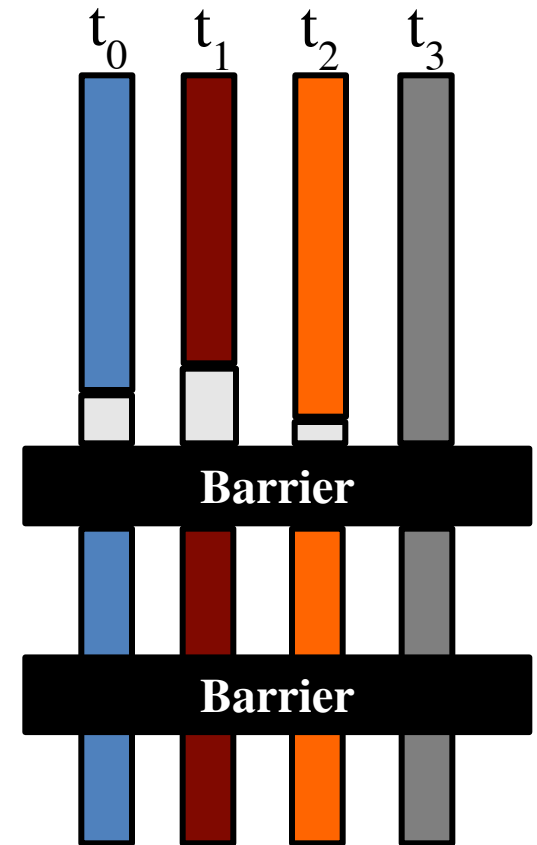
Global Synchronization Barrier

- At a barrier
 - All threads wait until all other threads have reached it
- Strawman implementation (**wrong!**)

```
global (shared) count : integer := P
```

```
procedure central_barrier
  if fetch_and_decrement(&count) == 1
    count := P
  else
    repeat until count == P
```

- What is wrong with the above code?



Sense-Reversing Barrier

- Correct barrier implementation:

```
global (shared) count : integer := P
global (shared) sense : Boolean := true
local (private) local_sense : Boolean := true
```

```
procedure central_barrier
    // each processor toggles its own sense
    local_sense := !local_sense
    if fetch_and_decrement(&count) == 1
        count := P
        // last processor toggles global sense
        sense := local_sense
    else
        repeat until sense == local_sense
```

Other Barrier Implementations

- Problem with centralized barrier
 - r All processors must increment each counter
 - r Each read/modify/write is a serialized coherence action
 - m Each one is a cache miss
 - r $O(n)$ if threads arrive simultaneously, slow for lots of processors
- Combining Tree Barrier
 - r Build a $\log_k(n)$ height tree of counters (one per cache block)
 - r Each thread coordinates with **k** other threads (by thread id)
 - r Last of the **k** processors, coordinates with next higher node in tree
 - r As many coordination address are used, misses are not serialized
 - r $O(\log n)$ in best case
- Static and more dynamic variants
 - r Tree-based arrival, tree-based or centralized release